Processes of Object-Oriented Software Development - Spring 2026

Personal Contact Manager Web Application

presentation

Team Members

Name	Description
Tales Souza Dias	Worked on some of the Contacts API endpoints and API documentation
Jashua Sauri Colon	Frontend Work on JS and HTML
Giovana Medeiros Castro	Worked on the database and created the ERD and Gantt Chart for the project.
Luiz S Dias Gomes	Project Manager
Maria Colmenares	Worked on the backend creating some of the methods for the API implementation
Joseph Ervin	Worked on the Contacts API endpoints and error handling

Technology used

Frontend

- HTML - Login, Register, Contact pages
- CSS - Custom layout + Tailwind CDN
- JavaScript - Client-side logic + jQuery

Backend

- PHP
- Slim framework

Database

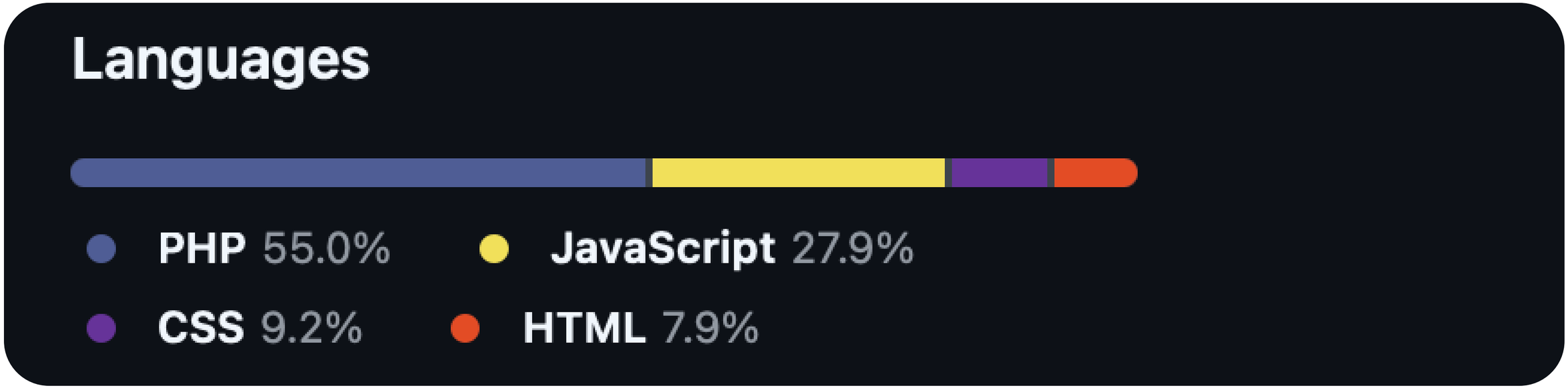
- SQLite - Development and API testing
- MySQL Workbench - Database modeling and ERD design
- Relational schema with Users and Contacts tables

Development Tools

- Composer
- GitHub - Version control and collaboration
- Discord - Communication

Project Successes

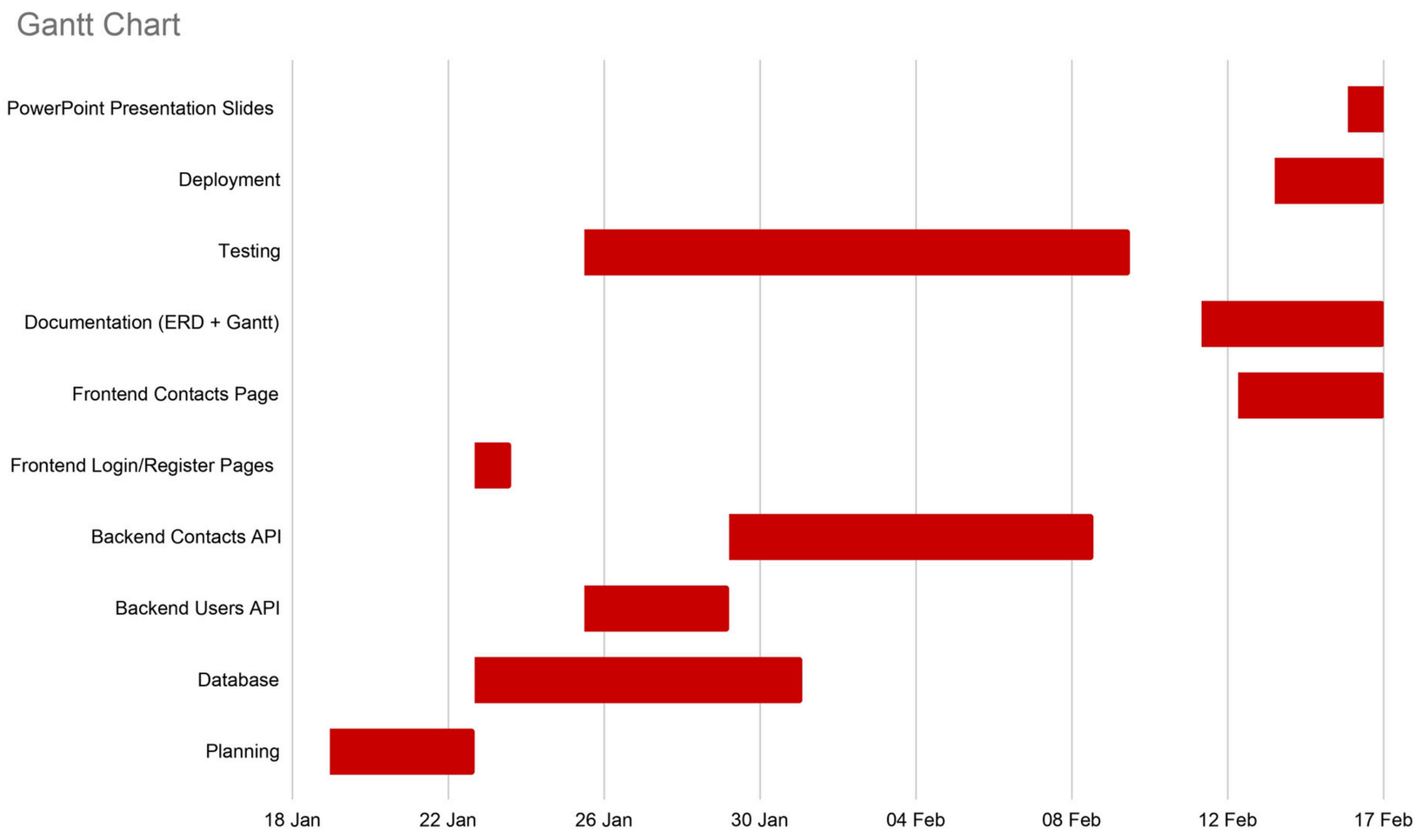
- Completed a working full-stack application.
- Successfully integrated frontend, backend, and database.
- Adapted to team changes and delivered on time.
- Team contributions on GitHub, with all changes merged only after review and approval.



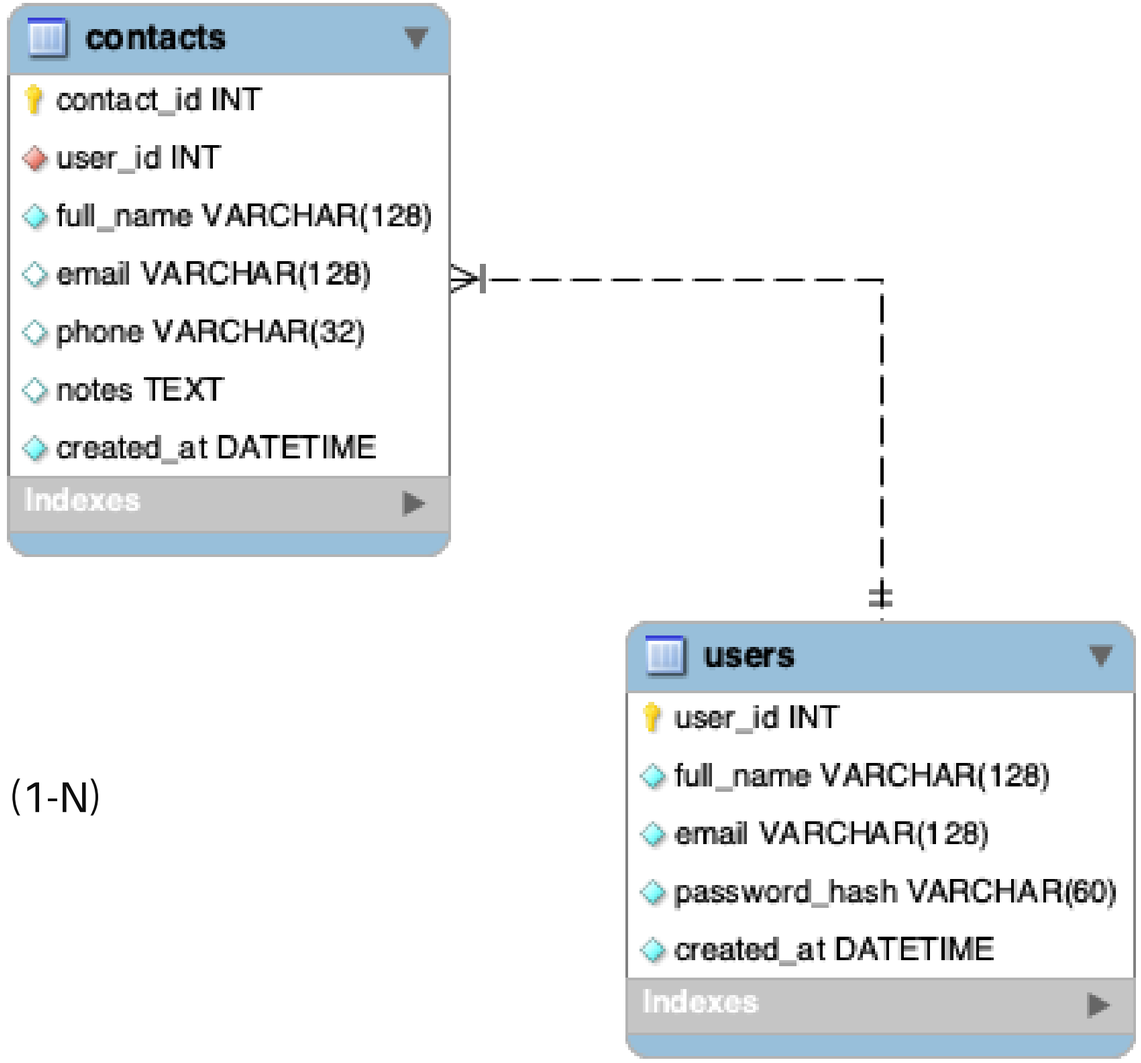
Project Challenges

- Switch server built-in Nginx to Apache
- Set up tests on GitHub
- 14 opened issues on GitHub. About 40 pull requests
- Set up the backend and the frontend to be serviced.
- DNS configuration

Gantt Chart


Gantt Chart
PowerPoint Presentation Slides
Deployment
Testing
Documentation (ERD + Gantt)
Frontend Contacts Page
Frontend Login/Register Pages
Backend Contacts API
Backend Users API
Database
Planning
18 Jan
22 Jan
26 Jan
30 Jan
04 Feb
08 Feb
12 Feb
17 Feb

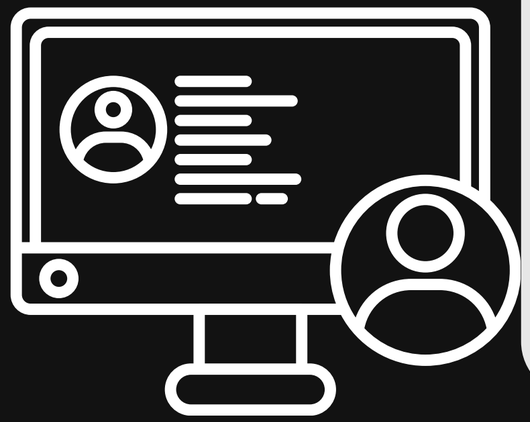
ERD



- Two main entities: Users and Contacts
- One-to-many relationship (One User → Many Contacts) (1-N)
- user_id used as a foreign key in Contacts

```
{
  "ok": true,
  "contacts": [
    {
      "contact_id": 1,
      "full_name": "JohnDoe2",
      "email": "johndoe@email.com",
      "phone": "4832904823",
      "notes": "test",
      "created_at": "2026-02-06 16:23:25"
    }
  ]
}
```

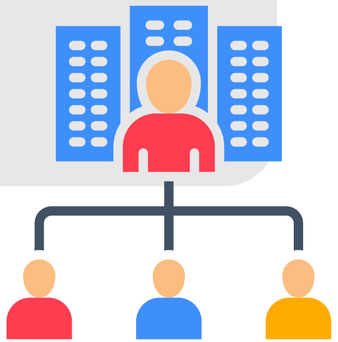

API demonstration

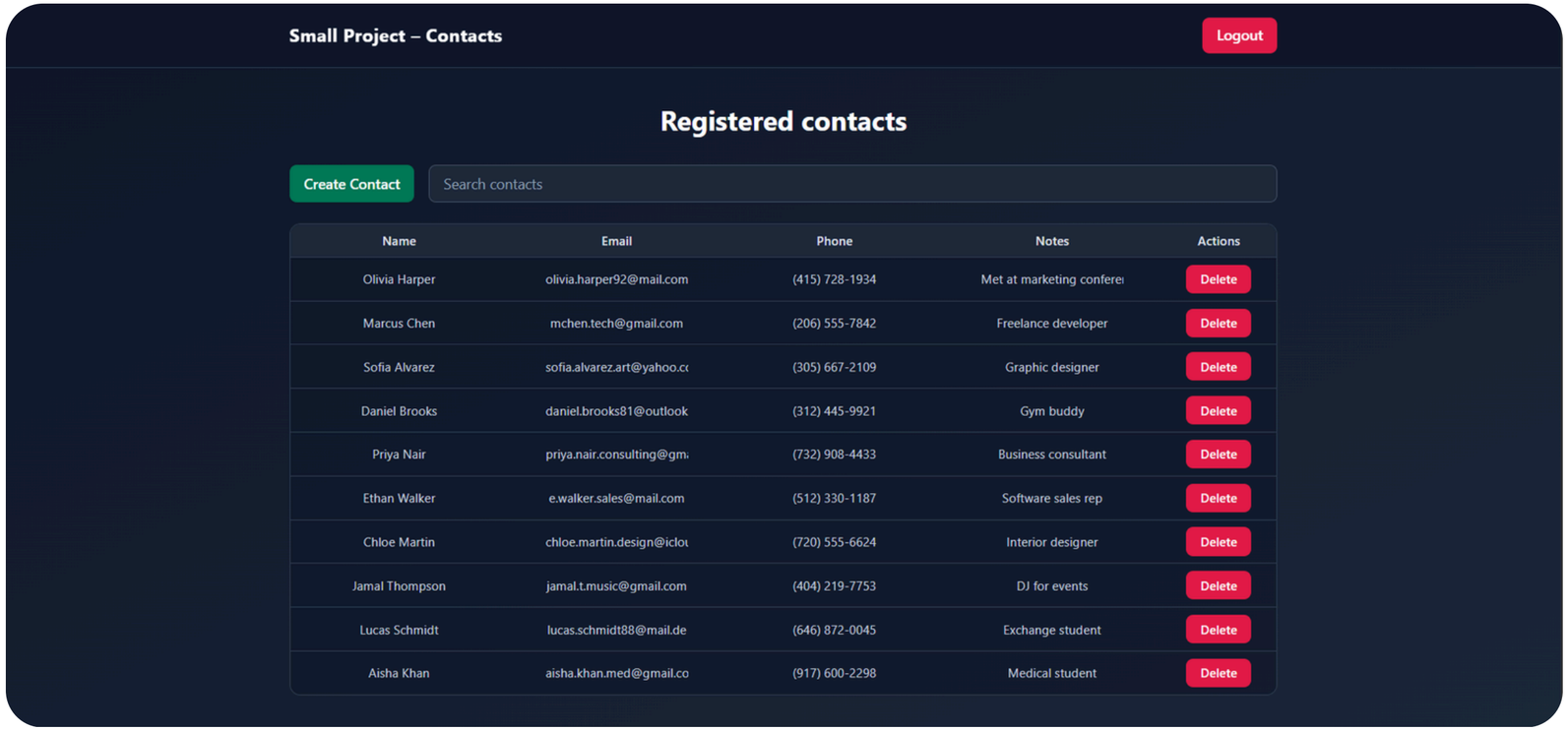


- Interactive API documentation using Swagger (OpenAPI 3.0)
- Demonstrates Contacts endpoint
- Shows request parameters and JSON responses
- Documentation in the web browser

Application demonstration

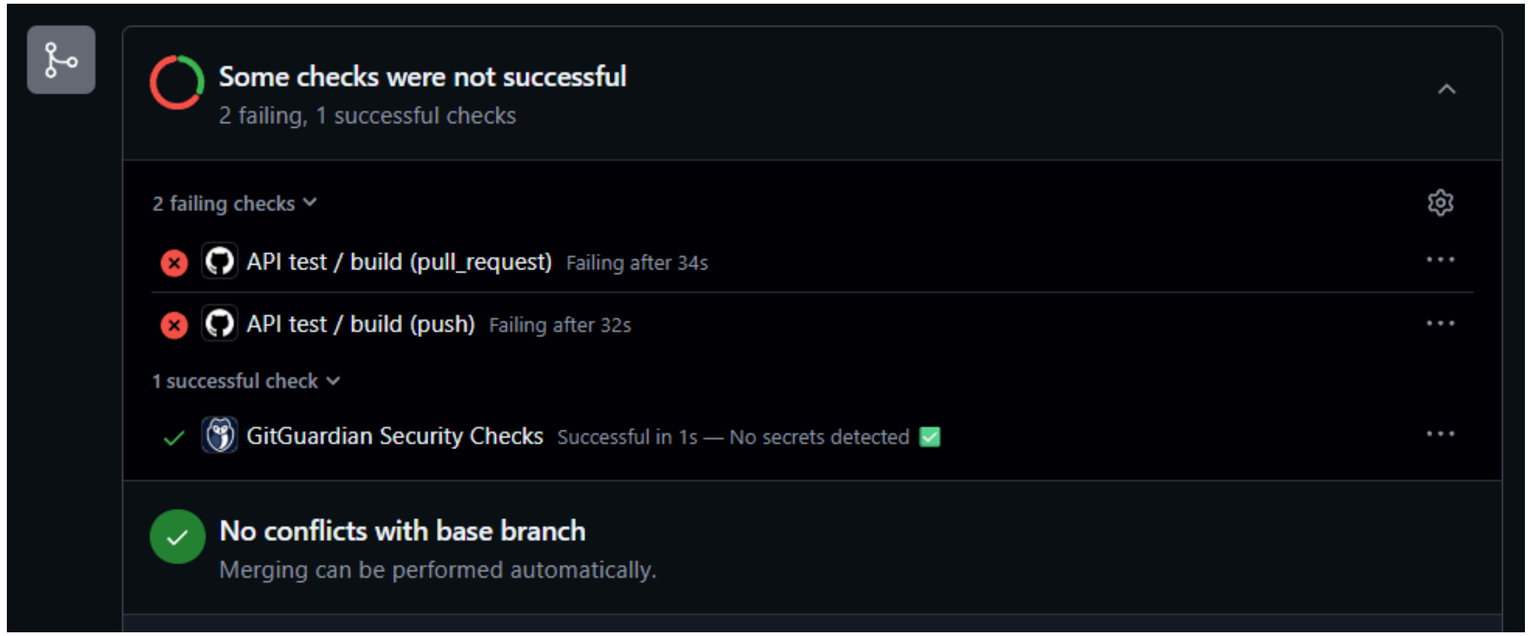
- User authentication (login required)
- View registered contacts
- Create new contacts
- Display stored contact information
- Delete existing contacts





Tests - GitHub actions

- Automatically rejects commits and pull requests if any tests fail
- Ensures previously working routes remain functional
- Prevents regressions from being introduced into the codebase
- Provides an extra layer of security when merging changes into the main branch



Questions ?